Lecture 8 &9

Methods of the determination of dietary requirements and standards
&
Nutrient Density

Determination of dietary adequacy

four guidelines have been developed to aid health professionals and consumers <u>choose diets to meet nutrient needs :</u>

1- Recommended dietary allowances .

2- Food group systems .

3- Food composition table .

4- Dietary guidelines for Americans .

5- Nutrient density .

Recommended dietary allowances

While planning balanced diets we need certain guidelines regarding the kinds and amounts of nutrients that we require for maintenance of good health . The RDAs is the guideline stating the amount of nutrients to be actually consumed in order to meet the requirements of the body. The RDAs is based on the requirements.

The requirements to perform specific functions in the body and to prevent deficiency symptoms .

Recommended dietary allowances

RDA should also maintain satisfactory stores of the nutrients in the body .

Recommended Dietary Allowances = Requirements + Margin of safety .

The margin of safety is added to take care of factors such as :

1- Losses during cooking and processing .

2- Short periods of deficient intake .

3- Nature of the diets.

4- Individuals variations in requirements.

RDAs depending on :

1- Gender .

2- Age .

3- Body size .

4- Activity level .

5- Special physiological state.

Disease and drugs prescribed for treatment can alter the requirement for one more nutrients .

RDAS FOR SPECIFIC NUTRIENT

1- The RDAs are expressed in metric units such as :

A- Kilocalorie (kCal) .

B- Grams (g)

C- Milligrams (mg).

D- Micrograms (μ g).

2- They are based on gender and activity levels such as : (sedentary, light, moderate, and heavy).

3- The RDAs for B- complex vitamins, B1, B2 and niacin are based on (kCal) or energy . The major role of these three vitamins is the release of energy from carbohydrates , proteins and fats .

RDAS FOR SPECIFIC NUTRIENT

The RDAs for B1, B2 , and niacin are (0.5 , 0.55 , 6.6 mg/ 1.000 kcal) respectively.

4- The RDAs for proteins are based on body weight. Adults need 1 g / kg body weight while infants, children, adolescents, and pregnant and lactating mothers need more protein to meet the demands of growth and body building .

5- The RDAs for infants are expressed per kg body weight .

RDAS FOR SPECIFIC NUTRIENT

6- RDAs for practically all nutrients increase during pregnancy and lactation to meet the needs of the growing foetus , and production of milk .

7- The RDAs for vitamin A are expressed in terms of retinol and β- carotene (provitamin A). β-carotene needs to be converted to vitamin A in the body .

(DRI)
Dietary Reference Intakes

Dietary Reference Intakes (DRI):

A set of reference values for energy and nutrients that can be used for planning and assessing diets for healthy people.

Dietary Reference Intakes

-Established by a committee of nutrition experts selected by the National Academy of Sciences (NAS)

-Based on latest scientific evidence regarding diet and health and health

-The first set, called the Recommended Dietary Allowances (RDA), was first published in 1941; revised ten times

-The series of DRI reports have been published since 1997

Dietary Reference Intakes (DRIs)

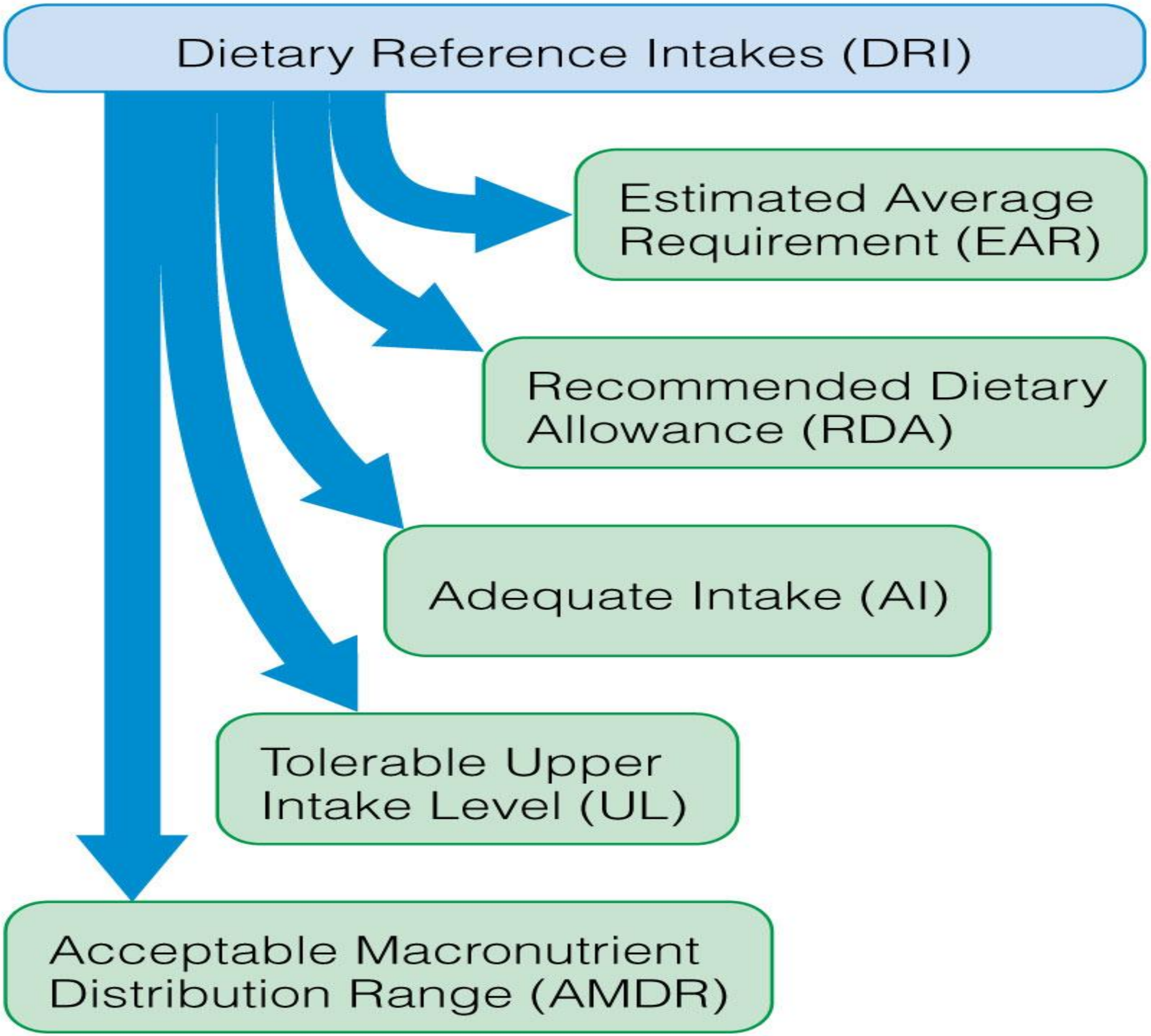
Focus on

- Maintaining good health
- Reducing the risk of developing chronic disease
- Avoiding unhealthy excess (toxic amounts of nutrients)

Establish nutrient recommendations for different life stages

- Age
- Gender
- Pregnant
- Lactating

Are periodically updated


Dietary Reference Intakes (DRI)
Estimated Average Requirement (EAR)
Recommended Dietary Allowance (RDA)
Adequate Intake (AI)
Tolerable Upper Intake Level (UL)
Acceptable Macronutrient Distribution Range (AMDR)

Dietary Reference Intakes (DRIs)

Include:

1-Estimated Average Requirement (EAR)

2-RDAs (Recommended Daily Allowances)

3-Adequate Intakes (AI)

4-Tolerable Upper Limit (UL)

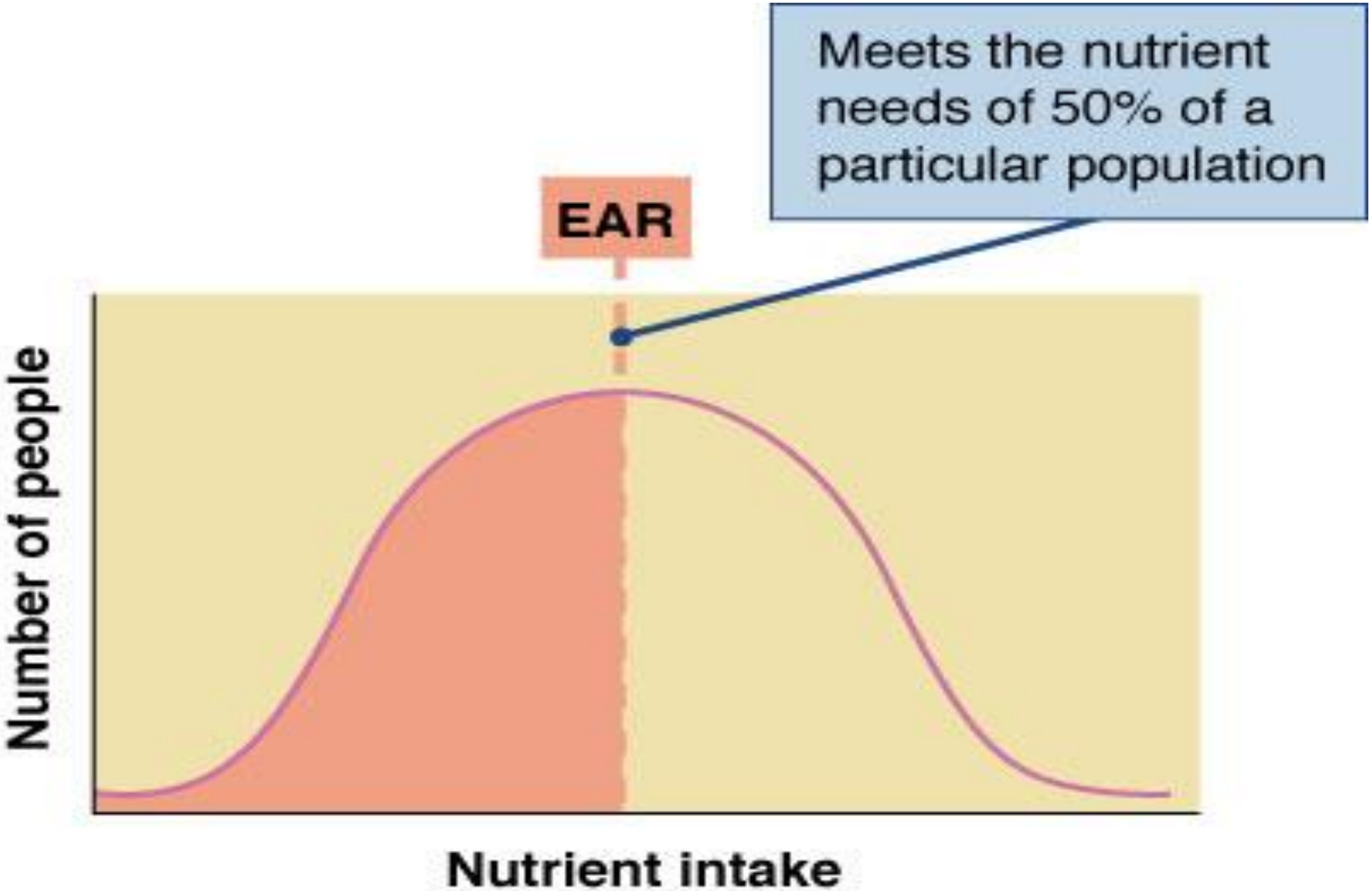
5-Acceptable Macronutrient Distribution Ranges (AMDR)

Reference Value Definitions

1-Estimated Average Requirement (EAR)

A daily nutrient intake value that is estimated to meet the requirements of half the healthy individuals in a group

Intake at which the risk of inadequacy is 0.5 (50percent) to an individual



ESTIMATED AVERAGE REQUIREMENT

Estimated Energy Requirements (EER)

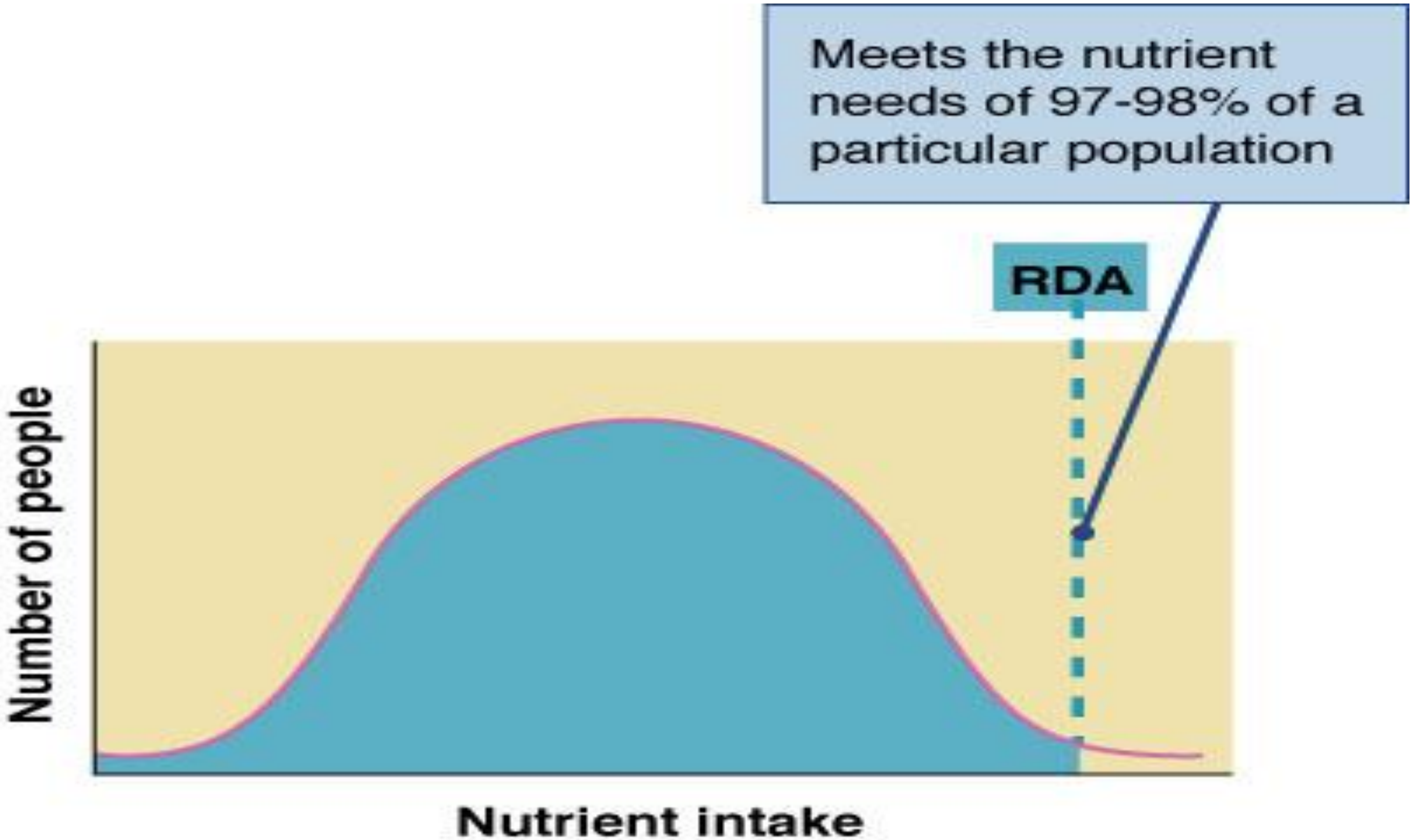
Amount of daily energy needed to maintain a healthy body and meet energy needs based on

- Age
- Gender
- Height
- Weight
- Activity level

2-Recommended Daily Allowance (RDA)

The average daily intake level that is sufficient to meet the nutrient requirement of nearly all(97-98%) healthy individuals in a particular life stage and gender group

The intake at which the risk of inadequacy is very small—only0.02to0.03(2to3 percent)



RECOMMENDED DIETARY ALLOWANCE

3-Adequate Intakes (AI)

- Set when there is not enough scientific evidence to determine an RDA

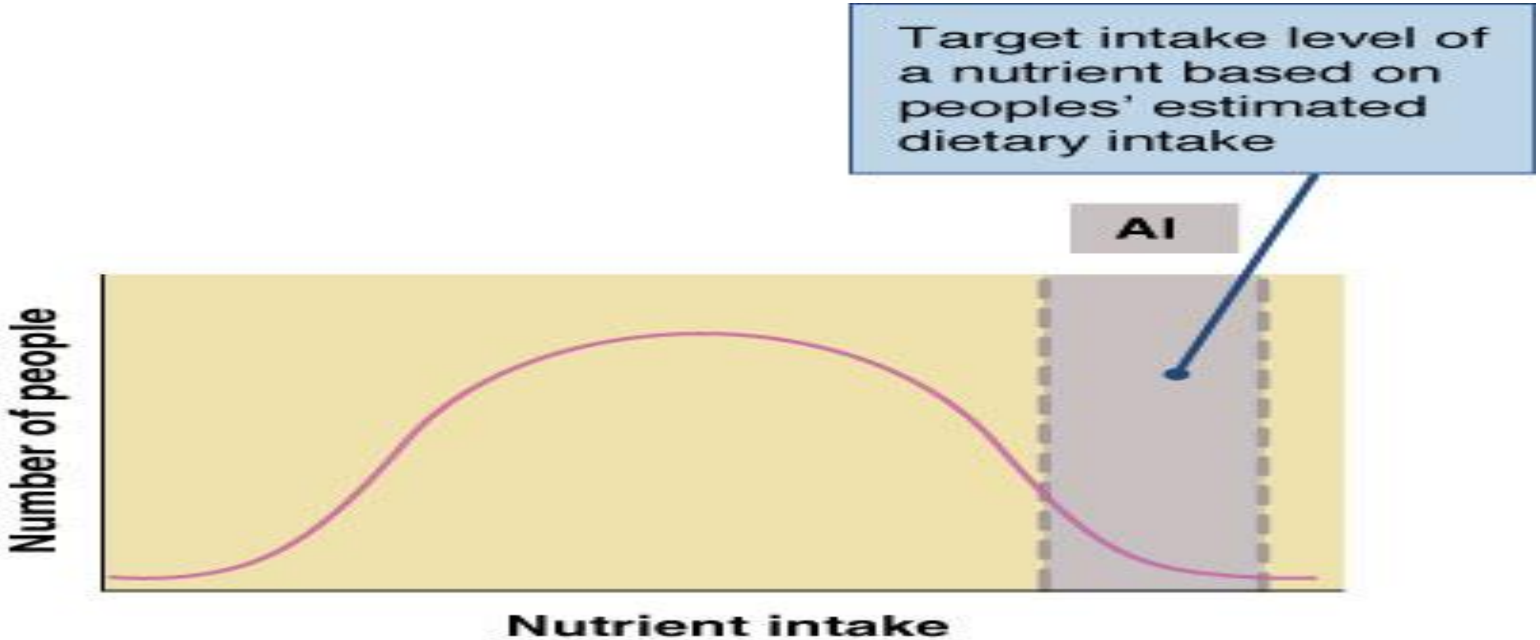
- Scientific estimate the amount of a nutrient that groups of similar individuals should consume to maintain good health

4-Tolerable Upper Limit (UL)

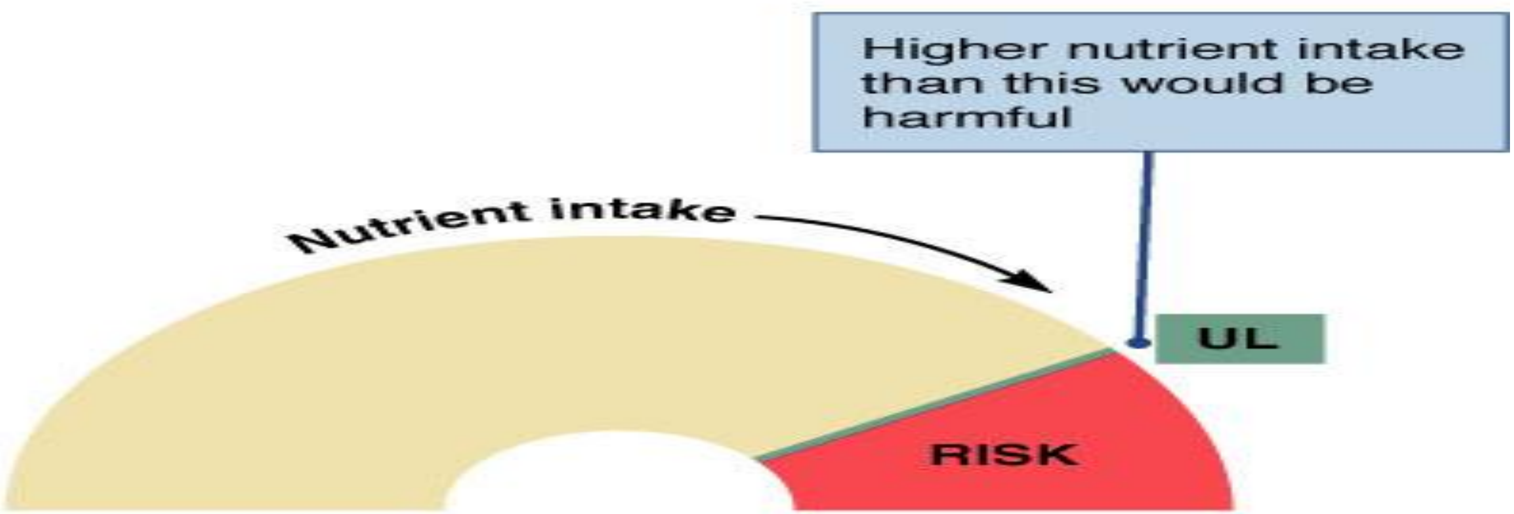
Highest level of a daily nutrient that is likely to pose no risk of adverse health effects to almost all healthy individuals

It is not intended to be a recommended level of intake.

The need for setting UL is the result of more and more people using large doses of nutrient supplements and the increasing availability of fortified foods.



ADEQUATE INTAKE



TOLERABLE UPPER INTAKE LEVEL

5-Acceptable Macronutrient Distribution Ranges (AMDR)

Addresses the recommended **balance** of fuel nutrients to meet physiological needs

- Carbohydrates 45–65% of daily kcal
- Fats 20–35% of daily kcal
- Proteins 10–35% of daily kcal

Consist of DRIs

The Daily Recommended Intake (DRIs)

consist of four components

1-Estimated Average Requirement (EAR)

2-Recommended Dietary Allowance (RDA)

3-Adequate Intake (AI)

4-Tolerable Upper Intake Level (UL)

Daily Value (%DV)

- Each type of reference value is calculated from daily intakes averaged over time (usually one or more weeks).
- Nutrition labeling
- The percent Daily Value (%DV) is based on 2000 Calorie diets and uses RDA or AI values.

EAR

If the sampling and end points are well – defined, the RDA can be calculated from the EAR.

RDA = EAR + 2 SD_{EAR} –

Where –

SD_{EAR} = standard deviation above the EAR •

Recommended dietary allowances

The RDAs is the guideline stating the amount of nutrients to be actually consumed in order to meet the requirements of the body. The RDAs is based on the requirements.

Estimation of Fluid Needs

Method	Weight	Fluid requirements
Weight	1-10 kg	100 ml/kg
	11-20 kg	50 ml/kg
	>20 kg (if < 50 yo)	20 ml/kg
	>20 kg (if > 50 yo)	15 ml/kg
Age	Infant	100 ml/kg
	Young active (16-30 yo)	40 ml/kg
	Young inactive (16-30 yo)	32 ml/kg
	Older (55-65)	30 ml/kg
	Elderly (65+)	25 ml/kg
Energy	1 ml / kcal.	
Fluid Intake	Urine output + 500 ml per day (pre-dialysis or odemia) Urine output +1000ml per day (dialysis)	

Calorie needs for Children and Adolescence (in Kraus)

Age (Yrs)	BEE (factor x weight)	Per day
0-0.5	108	650
0.5-1	98	850
1-3	102	1300
4-6	90	1800
7-10	70	2000
Male 10-14 15-18	 55 45	 2500 3000
Female 10-14 15-18	 47 40	 2200 2200

1- Energy Needs Based on weight and activity

Condition	Goal	Sedentary	Moderate	Active
Overweight	Weight loss	20 - 25 Kcal/day	30 Kcal/day	35 Kcal/day
Normal	Maintenance	30 Kcal/day	35 Kcal/day	40 Kcal/day
Underweight	Weight gain	30 Kcal/day	40 Kcal/day	40 - 50 Kcal/day

2- BEE according to Harris-Bendict equation

BEE men = 66+ (13.7 x weight in kilos) + (5 x height in cm) – (6.8 x age in years)

BEE women = 655+ (9.6 x weight in kilos) + (1.8 x height in cm) – (4.7 x age in years)

Total daily caloric needs = BEE x physical activity factor x Injury factor

Common Equivalents

1 cup (c) = 250 ml or g		1 kg = 1000 g	
1 ounce (oz) = 28 g		1 gm = 1000 mg	
1 teaspoon (tsp) = 5 g		1 mg = 1000 µg	
1 tablespoon (tbs) = 15 g		1 m = 100 cm	
1 inch (in) = 2.5 cm		F= (C *(9/5))+32	
1 foot (ft) = 30 cm		C= (F – 32) * (5/9)	
1 pound (Ib) = 454 g			
1 kg = 2.2 pound			
1 K cal = 4.2 K J			
1 gallon = 3.8 L			
1 Quart = (qt) = .95 L			